

Road Marking Segmentation Problem

Team: TechTitans

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Abstract—In autonomous navigation and intelligent traffic monitoring systems, precise road marker segmentation is essential, particularly when using Unmanned Aerial Vehicles (UAVs). But when it comes to top-down views of the scene, there is a domain gap because the majority of publicly available datasets were gathered from a ground-level perspective. The goal of this research is to use deep learning techniques to tackle the problem of segmenting road markers with different views and road conditions. First, the road marking dataset from CeyMo is used to train segmentation models like DeepLabV3+ and SegFormer. The collection includes many sorts of items, including road symbols, arrows, and lane markings. Metrics like Mean Intersection over Union (mIoU) are used to gauge the model's performance. The model can be more easily adapted to the UAV-based perspective dataset, such the AU Drone Dataset, by using the transfer learning approach with the ground-level perspective dataset, like CeyMo. The potential of using transformer-based and convolution-based models for semantic segmentation is presented in this work.

Index Terms—Semantic Segmentation, Deep Learning, Transfer Learning, Domain Adaptation, UAV, Road Marking Detection, SegFormer, DeepLabV3+

I. INTRODUCTION

RECOGNISING road markings is becoming more and more crucial as the demand for autonomous vehicles and sophisticated traffic management grows. These include lane lines, arrows, and pedestrian crossings. Vehicles use the important information carried by these marks. Road markers have thus far been identified using computer vision. However, various types of road surfaces, occlusions, and lighting conditions bring up great challenges in mark segmentation and identification. Deep learning models have recently shown notable advances, particularly in the field of semantic segmentation. The ability to comprehend traffic scenes has greatly enhanced, thanks to these models. The primary issue that has been found is that models that are trained on ground-level images do not function as intended when applied to aerial photos taken by drones. This project's primary objective is to make sure that models are first trained on a conventional picture set before being enhanced to function with drone-based image sets, utilising transfer learning techniques.

II. LITERATURE REVIEW

Semantic segmentation algorithms have come a long way since the days when they could only identify edges; today,

these algorithms utilize advanced machine learning techniques that can consider various factors within an environment. Here are some examples of literature that form the basis of our approach:

1. DeepLab: Semantic Image Segmentation with Deep Convolutional Nets, Atrous Convolution, and Fully Connected CRFs This work made two significant contributions to dense prediction tasks:

Atrous convolution: This technique gives the network explicit control over feature response resolution and expands the field of view without increasing computation or parameters.

Atrous spatial pyramid pooling (ASPP): This technique enables multiscale object segmentation by sampling the input feature layer using different rate filters. This is particularly important because road markings will appear at different resolutions depending on camera angle.

2. CeyMo: Road Marking Detection Benchmarking Datasets have an important influence on the efficiency of a model. CeyMo is the dataset designed to resolve the problem of insufficient data for road markings.

It offers an innovative benchmark for 11 categories of road markings in tough situations, including darkness, rain, and glare.

The study reveals that although conventional models work effectively in case of high contrast markings, there is still a considerable "domain gap" between models trained on ground data and those tested on aerial images or UAVs.

3. SegFormer: Efficient and Global Context-aware

SegFormer is one step towards Transformer-based neural networks for semantic segmentation tasks.

Position Encodings Not Required: The unique feature of SegFormer lies in its lack of position encodings, which enables it to perform well despite there being a mismatch between test and training resolutions.

Hierarchical Encoder: The model is built on a hierarchical architecture capable of extracting both low-resolution and high-resolution features simultaneously, allowing it to be very effective at understanding global context throughout an image, especially long lane lines.

4. Feature Fusion and High-Precision Detection The latest studies from 2024 point out the necessity of utilizing the

Feature Fusion approaches for enhancing the precision level of non-linear objects.

With the help of designing unified models that integrate various features at different scales, scientists managed to obtain much higher mIoU values in complex databases such as ApolloScape.

The study shows the need for combining several kinds of traffic marking detection (not only lane lines) to receive complete right-of-way information.

III. METHODOLOGY

A. Data Description

For this project, we used the Ceymo road marking dataset for training and evaluation. It has diverse road scenes with annotated masks that categorize various road marking types, including lane lines, arrows, crossings, and symbols.

The dataset is divided into three subsets:

- **Training set:** It is Used for model weight optimization.
- **Validation set:** It is Used to tune hyperparameters and monitor performance.
- **Test set:** It is Used to evaluate the models.

Every image has a ground-truth mask in which every pixel value represents a specific class label. This helps the model to perform dense pixel level classification which is the main objective of semantic segmentation.

B. Data Preprocessing

For data consistency and fast convergence, we did a few preprocessing steps:

- **Resizing:** Scaled all images and masks to a fixed resolution which is compatible with the model.
- **Normalization:** Pixel intensities were scaled (to a range of $[0, 1]$ or standardized using ImageNet means) to stabilize the training process.
- **Mask Encoding:** Annotation files were converted into integer encoded segmentation masks, where each unique value corresponds to a specific road marking category.
- **Data Loading:** data pipelines were made using PyTorch `DataLoader` to do batch processing and asynchronous data fetching.

C. Model Implementation

We implemented and compared two architecturally different deep learning models to evaluate their accuracy in road marking detection.

1) *DeepLabV3+*: DeepLabV3+ is a state of the art convolutional neural network (CNN) designed for semantic segmentation. It is good at capturing multi-scale contextual information with the use of Atrous Spatial Pyramid Pooling (ASPP).

- **Encoder:** Uses a pre-trained backbone (e.g., ResNet) to extract visual features.
- **Decoder:** A dedicated decoder module refines object boundaries by fusing low level features with high level outputs.
- **Strengths:** Highly effective at capturing structured patterns and fine details in lane markings.

2) *SegFormer*: SegFormer is a transformer-based framework that removes the need for positional encodings and uses a hierarchical Transformer encoder with a lightweight All-MLP decoder.

- **Global Context:** Unlike CNNs, the self-attention mechanism helps the model to capture global dependencies across the entire image.
- **Efficiency:** SegFormer is designed to be more computationally efficient and faster than traditional convolution models.
- **Adaptability:** The attention based approach is robust in complex scenes where road markings occluded or vary significantly in shape.

D. Training Setup

Both models were trained with a combined pipeline within the MMSegmentation framework:

- **Environment:** Google Colab utilizing NVIDIA GPU acceleration.
- **Loss Function:** Cross-Entropy Loss, formulated as:

$$L = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log(\hat{y}_{i,c})$$

- **Optimizer:** Adam optimizer for adaptive management of learning rate.
- **Hyperparameters:** Models were trained for multiple epochs with early stopping based on validation mIoU to prevent overfitting.

E. Evaluation Metrics

To assess performance, we used the following metrics:

- **Mean Intersection over Union (mIoU):** This is for segmentation and it is calculated as the average overlap between the predicted masks and ground truth.
- **Pixel Accuracy:** The ratio of correctly classified pixels to the total number of pixels.
- **Loss Value:** The convergence behavior of the Cross-Entropy function during the training and validation parts.

F. Model Ensembling

In this process, we combined the performance of both the models to leverage the mIoU score. This was performed using by extracting the raw probabilistic maps also known as the logits from both the models.

This logits are then weighted to produce a final refined prediction mask, reducing the losses of both the models to prevent false negatives.

- **Preprocessing:** The preprocessing part consists of standardizing the input by resizing it to 512x512 pixels.
- **Loading Checkpoints:** Best weights from the training of segformer and deeplabv3+ are imported here.
- **Performance Evaluation:** The performance of both the models are evaluated and compared.

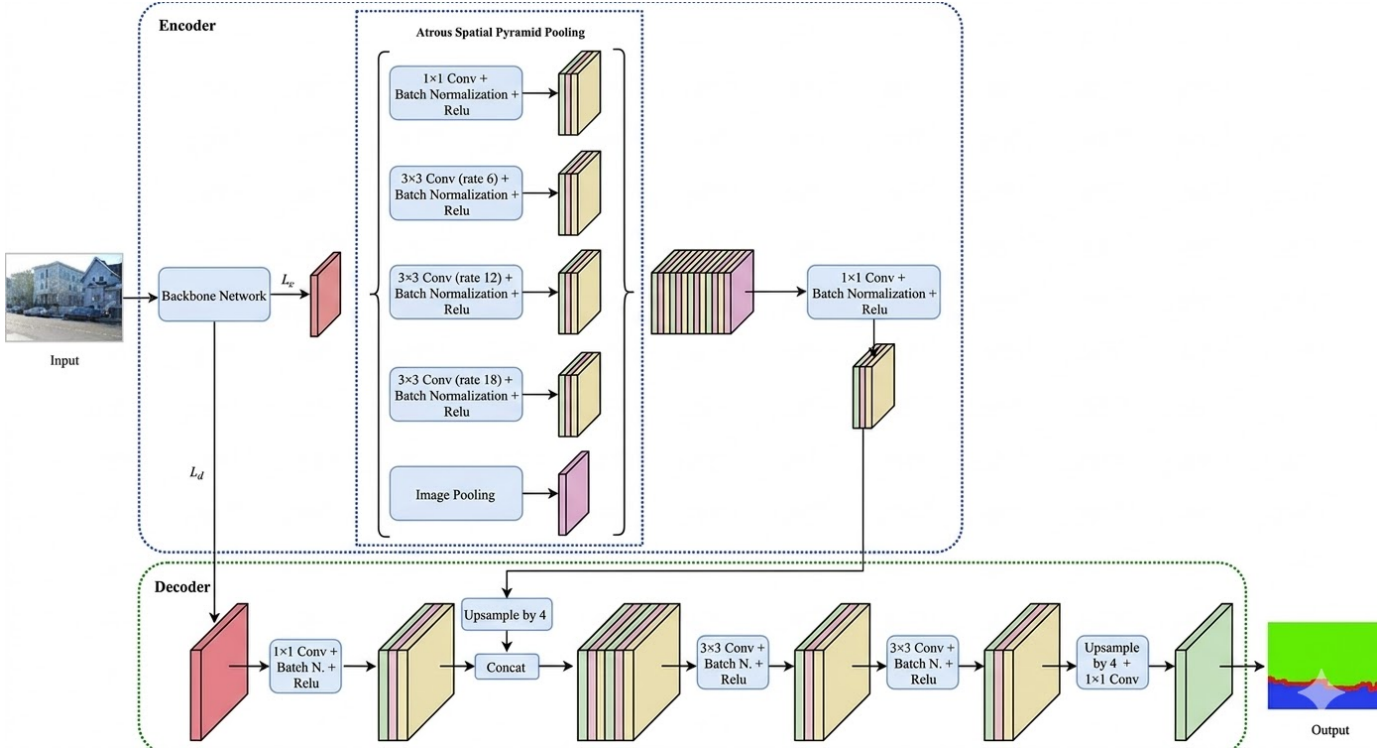


Fig. 1. Architecture of the DeepLabV3+ model showing the encoder-decoder structure and ASPP module.

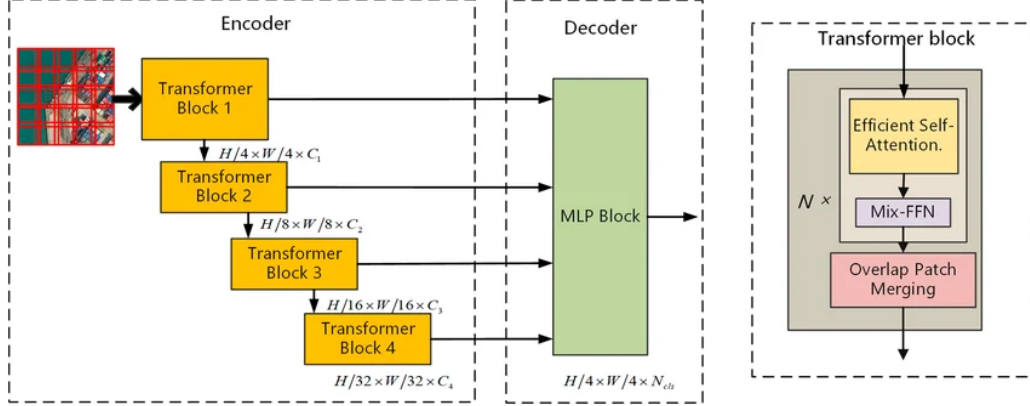


Fig. 2. Segformer Encoder-Decoder Architecture

IV. RESULTS AND EVALUATION

In this project we have used two deep learning models for the semantic segmentation of road markings using drone imagery. The aim was to accurately classify pixels into 13 road marking categories (e.g., crossings, arrows, lanes) and a background class.

A. SegFormer Results

The baseline SegFormer model (without any finetuning) and finetuned model results have been mentioned in the table below.

The figure below (Fig. 3) shows the performance of the SegFormer model by comparing the input RGB images, the

ground-truth annotations, and the model's predicted semantic masks. The visualization shows the model's high precision in identifying high-contrast features like crosswalks and yellow lane markings and also showing difficulties with smaller or more complex symbols like arrows and bicycle icons. By

TABLE I
SEGFORMER PRE AND POST FINE-TUNING RESULTS

Model State	Mean IoU
Baseline (Zero-shot)	0.0020
Fine-tuned (Proposed)	0.5341
Improvement	+0.5321

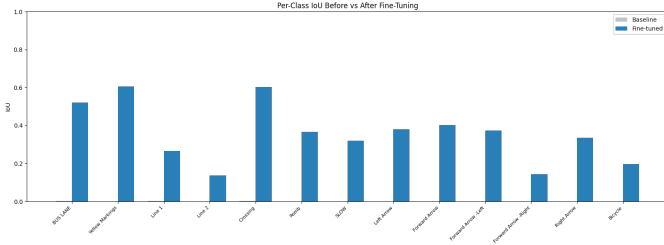


Fig. 3. Per Class Before vs After Fine-tuning

mapping these predictions back to the 14-class color scheme, the diagram is a critical diagnostic tool, showing that the model successfully generalizes spatial features even though there exist class imbalance in the dataset.

TABLE II
CLASS-WISE IOU RESULTS FOR SEGFORMER (RESNET-101)

Class	IoU Score
Background	0.0000
BUS LANE	0.7777
Yellow Markings	0.8927
Line 1	0.5893
Line 2	0.2386
Crossing	0.7425
Romb	0.5913
SLOW	0.5586
Left Arrow	0.4945
Forward Arrow	0.6615
Forward Arrow (Left)	0.4918
Forward Arrow (Right)	0.3193
Right Arrow	0.5854
Bicycle	0.0003
mIoU (Foreground)	0.5341

The SegFormer model shows high accuracy in identifying large scale and high contrast features, such as the "Yellow Markings" (0.8927), and "Crossing" (0.7425) classes. But the model shows lower precision for classes with thin geometries or lower frequency in the dataset, such as "Line 2" (0.2386) and "Forward Arrow (Right)" (0.3193). The imbalance between the "Forward Arrow" (0.6615) and its directional variants shows that while the model recognizes the general shape of arrows, it sometimes struggles with fine-grained orientation differences. Overall, the foreground mean IoU of 0.5341 confirms that the model can be used as a competitive baseline for autonomous navigation tasks.

B. DeepLabV3+ (ResNet-101 Backbone) Results

The DeepLabV3+ model uses a ResNet-101 backbone. It was evaluated in its pre-trained state and after fine tuning on the road marking dataset. As shown in Table III, fine tuning showed an increase in the overall Mean IoU (mIoU).

Class performance comparison is visualized in Figure 4, which shows the incremental gains that happened with transfer learning in various road marking categories.

The detailed class-wise IoU scores for the fine-tuned DeepLabV3+ model are presented in Table IV.

TABLE III
DEEPLABV3+ mIoU: PRE-TRAINED VS. FINE-TUNED

Model State	Mean IoU (mIoU)
Pre-trained (Baseline)	0.6605
Fine-tuned (Proposed)	0.6716
Improvement	+0.0111

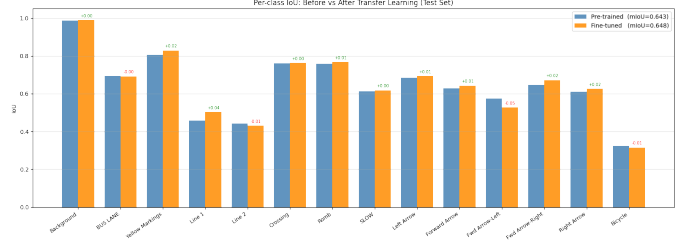


Fig. 4. Per-class IoU: Before vs. After Transfer Learning for DeepLabV3+

The DeepLabV3+ architecture shows a higher baseline and fine-tuned performance in comparison to SegFormer on this specific dataset. classes like **Yellow Markings** (0.814) and **Crossing** (0.771) show robust segmentation. While the **Bicycle** class (0.266) remains the most challenging due to its complex geometry and smaller pixel footprint, the model maintains a high degree of consistency across directional arrows, suggesting that the Atrous Spatial Pyramid Pooling (ASPP) in DeepLabV3+ is effective at capturing multi-scale context for road signage.

Later, on performing the ensembling method (combination of both the models) we inferred the results, showcasing the improved classwise IoU values. The evaluation focuses on Intersection over Union (IoU) to determine spatial accuracy. Although the SegFormer algorithm performs outstandingly in "Yellow Markings" and "BUS LANE" categories, DeepLab is more robust for most directional arrow categories and "background". Interestingly enough, the Ensemble model shows insignificant improvements in some categories while perform-

TABLE IV
CLASS-WISE IOU RESULTS FOR DEEPLABV3+ (FINE-TUNED)

Class	IoU Score
Background	0.990
BUS LANE	0.724
Yellow Markings	0.814
Line 1	0.517
Line 2	0.443
Crossing	0.771
Romb	0.782
SLOW	0.645
Left Arrow	0.731
Forward Arrow	0.702
Fwd Arrow-Left	0.678
Fwd Arrow-Right	0.682
Right Arrow	0.659
Bicycle	0.266
Mean IoU (mIoU)	0.6716

ing poorly in "Forward Arrow - Left" category. Overall, the Test mIoU for Ensemble is 0.6628, which is still below the maximum Test mIoU of 0.6716 for DeepLab. Detailed performance have been showcased in table V.

V. DISCUSSIONS

The outcomes show how effective deep learning models are at semantic segmentation for traffic signs. According to the results, transformer-based models—like SegFormer—have been found to perform better when handling spatial relationships than conventional CNN-based models. Nonetheless, it is necessary to talk about several limitations: This work's data set is ground-based rather than UAV-based. For aerial views, the models employed in this work might not function properly. The outcomes are directly impacted by the quality of the annotation utilised in the work. A significant problem that will be addressed in the future through transfer learning is the domain gap in the datasets.

VI. CONCLUSION

The project has shown how deep learning techniques can be used to segregate road markers. A multi-class road marking dataset was utilised to evaluate the models, DeepLabV3+ and SegFormer. The outcomes demonstrated that the ensembling method which is the combination of transformer-based and cnn-based models performed better in segmentation tasks. Future research will focus on improvising the model performance and adapting the models on UAV-based datasets using transfer learning. After that, the models will be capable of handling various situations and viewpoints. The study contributes to the advancement of self-driving cars.

VII. CONTRITBUTION

A. Megh Patel

Ensemble & Comparative Analysis

- Metric Evaluation: Conducted comparative analysis using mIoU, Pixel Accuracy, and Cross-Entropy Loss.
- Synthesis: Compared the geometric precision of CNNs against the spatial relationship handling of Transformers.
- Future Roadmap: Defined the framework for future UAV-based transfer learning and autonomous car integration.

B. Devanshi Pathak

DeepLabV3+ Implementation

- CNN Architecture: Implemented DeepLabV3+ with a ResNet-101 backbone for dense pixel classification
- Multi-Scale Detection: Leveraged Atrous Spatial Pyramid Pooling (ASPP) to capture road markings of varying sizes
- Performance: Achieved the project's highest mIoU of 0.648 through optimized training

C. Konark Karia

Literature Review & Project Strategy

- Research Foundation: Documented the shift from traditional vision to deep learning and identified the CeyMo benchmark
- Strategic Planning: Formulated the methodology to address the "domain gap" between ground-level and UAV views
- Data Pipeline: Established preprocessing steps, including normalization and integer mask encoding

D. Dharmi Vekariya

SegFormer Implementation

- Transformer Framework: Implemented the SegFormer model using a hierarchical encoder and All-MLP decoder
- Fine-Tuning: Conducted experimental runs that improved the baseline mIoU from 0.0013 to 0.3576
- Spatial Analysis: Analyzed how self-attention captures global dependencies in high-contrast features like cross-walks

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TABLE V
PERFORMANCE COMPARISON OF SEGFORMER, DEEPLAB, AND ENSEMBLE MODELS

Class	SegFormer	DeepLab	Ensemble	Δ vs best
background	0.0000	0.9898	0.9907	+0.0009 (DL)
BUS LANE	0.7777	0.7236	0.7780	+0.0003 (SF)
Yellow Markings	0.8926	0.8142	0.8774	-0.0152 (SF)
Line 1	0.5895	0.5166	0.5132	-0.0763 (SF)
Line 2	0.2385	0.4426	0.4349	-0.0077 (DL)
Crossing	0.7429	0.7708	0.7708	+0.0000 (DL)
Romb	0.5913	0.7821	0.7821	+0.0000 (DL)
SLOW	0.5584	0.6452	0.6456	+0.0004 (DL)
Left Arrow	0.4944	0.7310	0.7421	+0.0112 (DL)
Forward Arrow	0.6614	0.7021	0.7087	+0.0067 (DL)
Forward Arrow - Left	0.4917	0.6776	0.4205	-0.2572 (SF)
Forward Arrow - Right	0.3192	0.6816	0.6833	+0.0017 (DL)
Right Arrow	0.5852	0.6592	0.6658	+0.0066 (DL)
Bicycle	0.0003	0.2663	0.2658	-0.0005 (DL)
Test mIoU	0.4959	0.6716	0.6628	-0.0088